NYC Nightlife Pulse

City Live Monitor: An End-to-End Hybrid Data Platform Blueprint

Data Engineering 2025/2026 | Assignment 2 (Group)

Gabriel Silva (113786) | Hugo Castro (113889)



Observing the Nocturnal City

By fusing batch history, periodic schedules, and streaming noise data, the platform creates a multidimensional model of urban nightlife dynamics.

Layer 3: Real-Time Signal

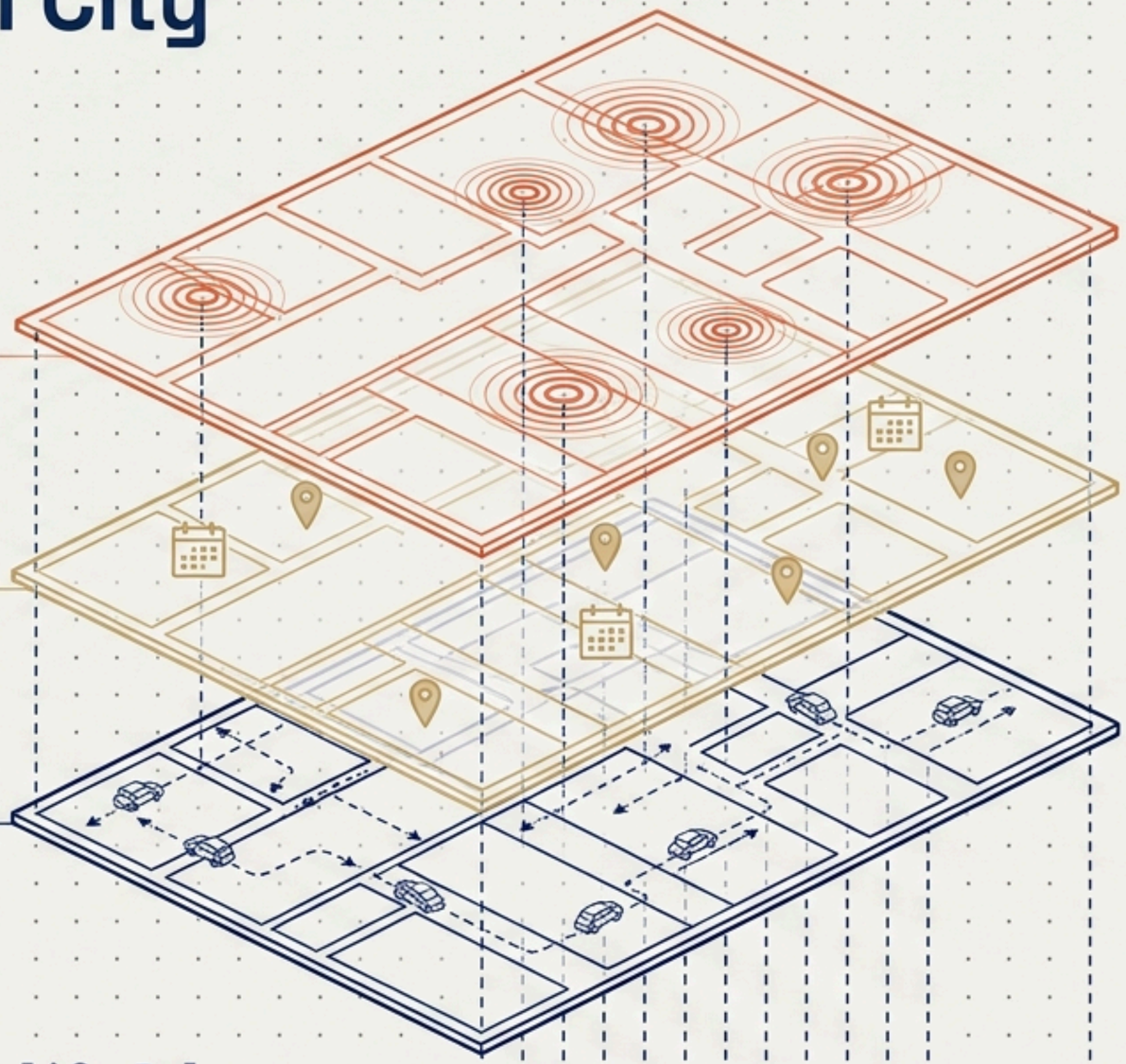
NYC 311 Noise Complaints - The emergent, almost real-time validation of human activity and friction.

Layer 2: Planned Agenda

NYC Permitted Events - The scheduled urban context explaining demand variations.

Layer 1: Historic Mobility

NYC TLC Trip Record Data - The baseline of nighttime movement and demand.



Nightlife Pulse

Translating Objectives into Engineering Rigor

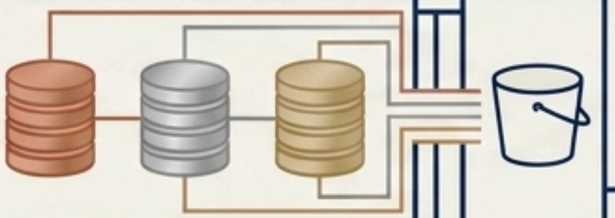
Academic Objectives

- Pipeline modularity
- Heterogeneous source integration
- Modern transformations
- Interactive visualization
- Reproducibility & reliability

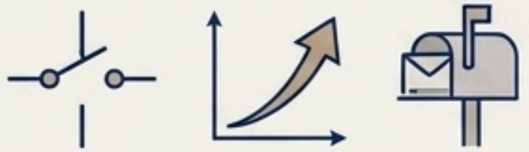
The Build

Decoupled Batch & Stateful Streaming pipelines

Medallion Architecture (Bronze/Silver/Gold) persisting to object storage



Robust fault tolerance (circuit breakers, exponential backoff, DLQs)



Containerized, heavily tested, and orchestrated local ecosystem



The Medallion Storage Layer



Bronze (Immutable Raw)

Append-only snapshots. Raw Parquet (Batch) and NDJSON (Stream). Secures exact historical state.

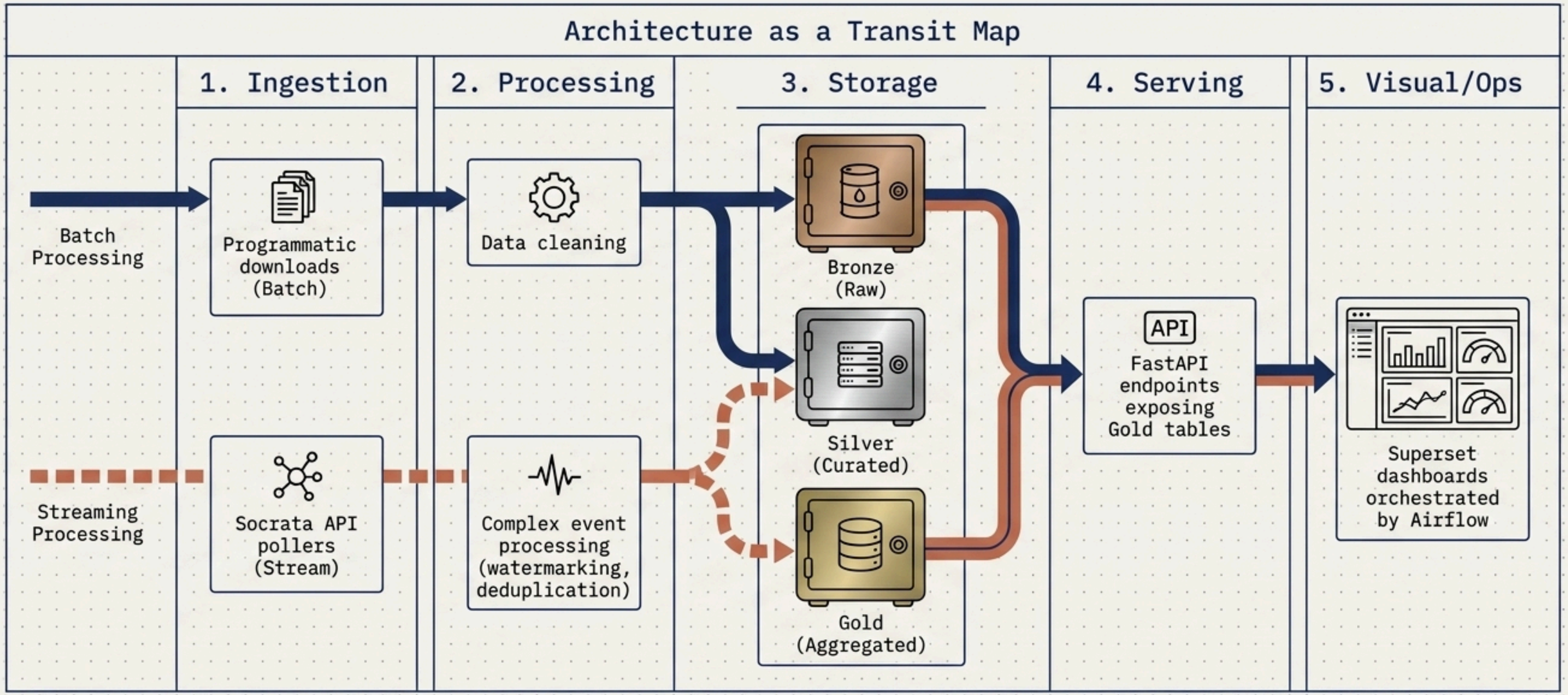
Silver (Curated Domain)

Cleaned, normalized data partitioned functionally by year/month. Standardized schema for analytical modeling.

Gold (Aggregated Analytics)

Canonically persisted Parquet tables optimized for direct query access (`taxi_demand_by_zone`, `event_schedule`, `noise_by_borough_hour`, `stream_metrics_5m`).

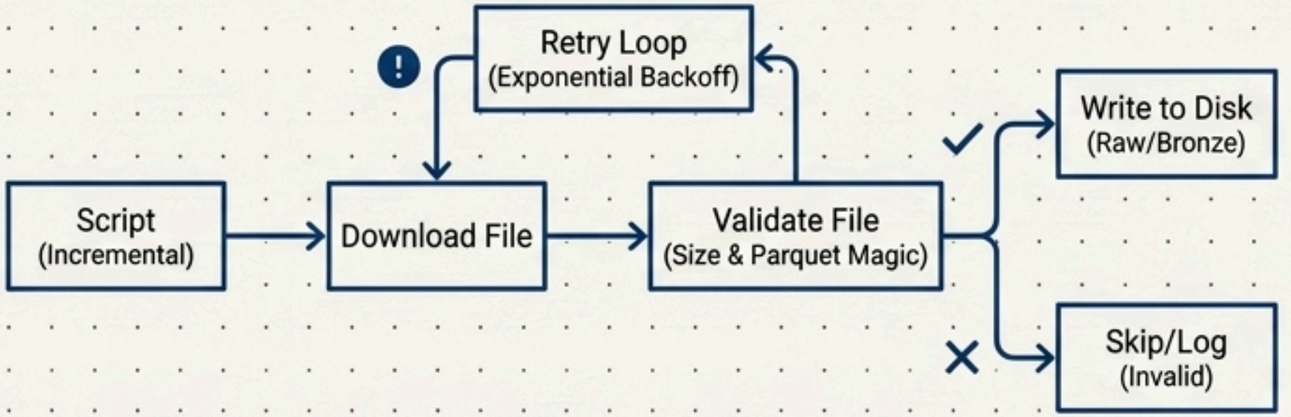
Macro Architecture: The 5-Pillar Blueprint



Ingestion Layer Dynamics: Fault-Tolerant Collection

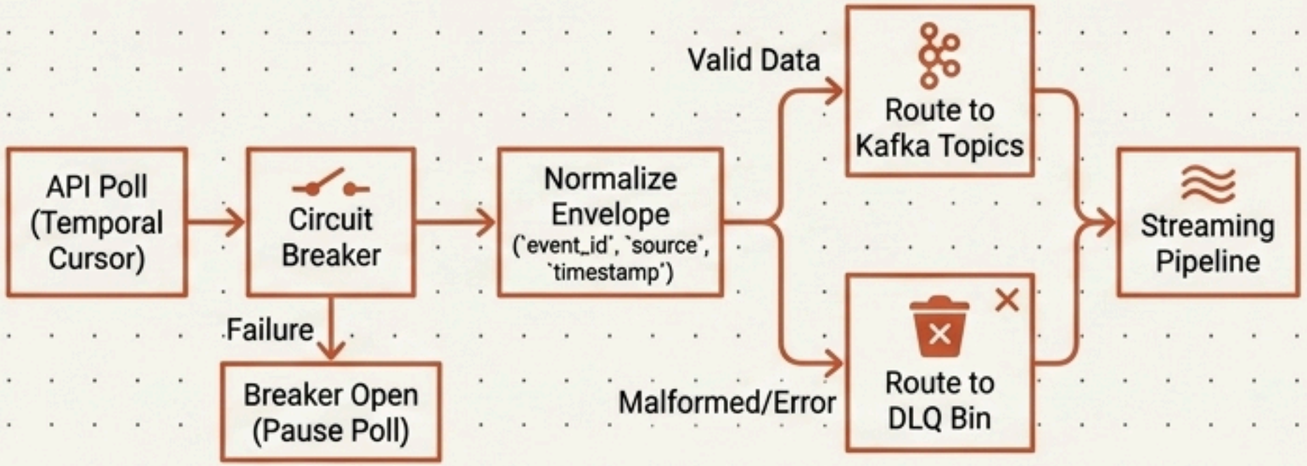
Batch Strategies - TLC

- Incremental logic (skips valid files)
- `ThreadPoolExecutor` for parallelization
- Retries mapped with exponential backoff
- File validation (minimum size + Parquet magic bytes)

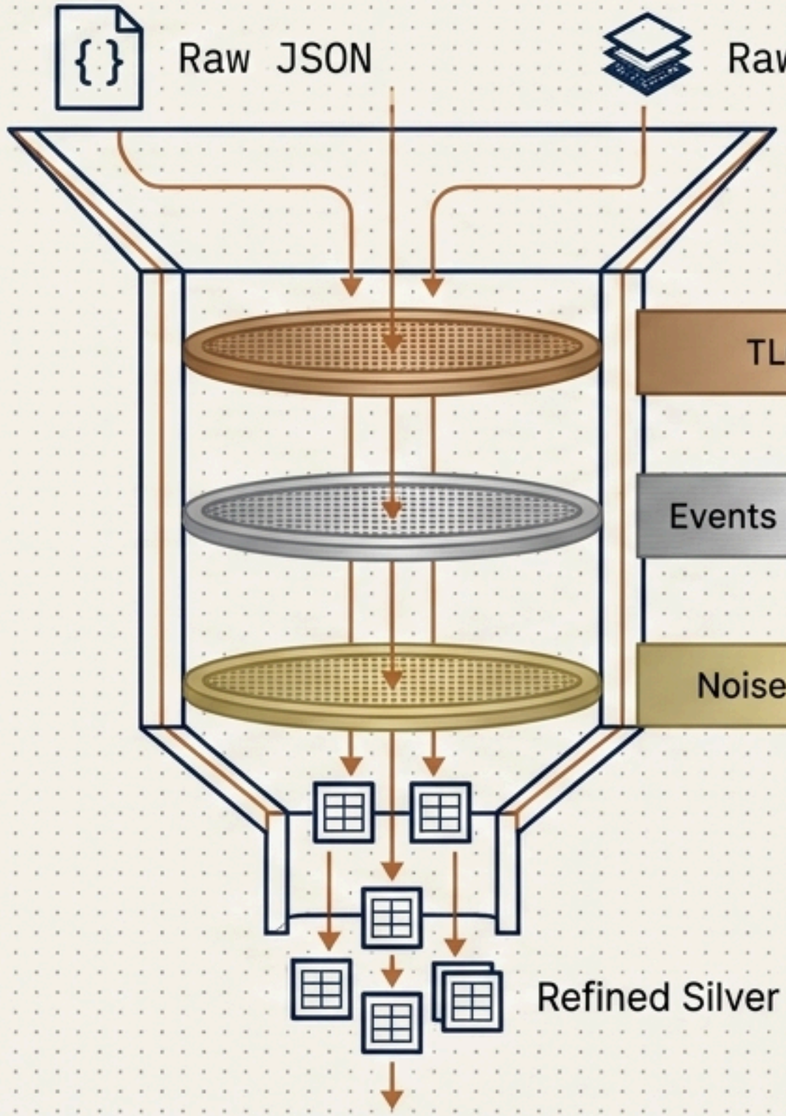


Streaming Strategies - Events & 311

- Incremental polling via persistent temporal cursors
- Common envelope normalization (`event_id`, `source`, `timestamp`)
- Circuit breakers for API resilience
- Dead Letter Queues (DLQ) routing for malformed events



Processing: Refining the Silver Layer

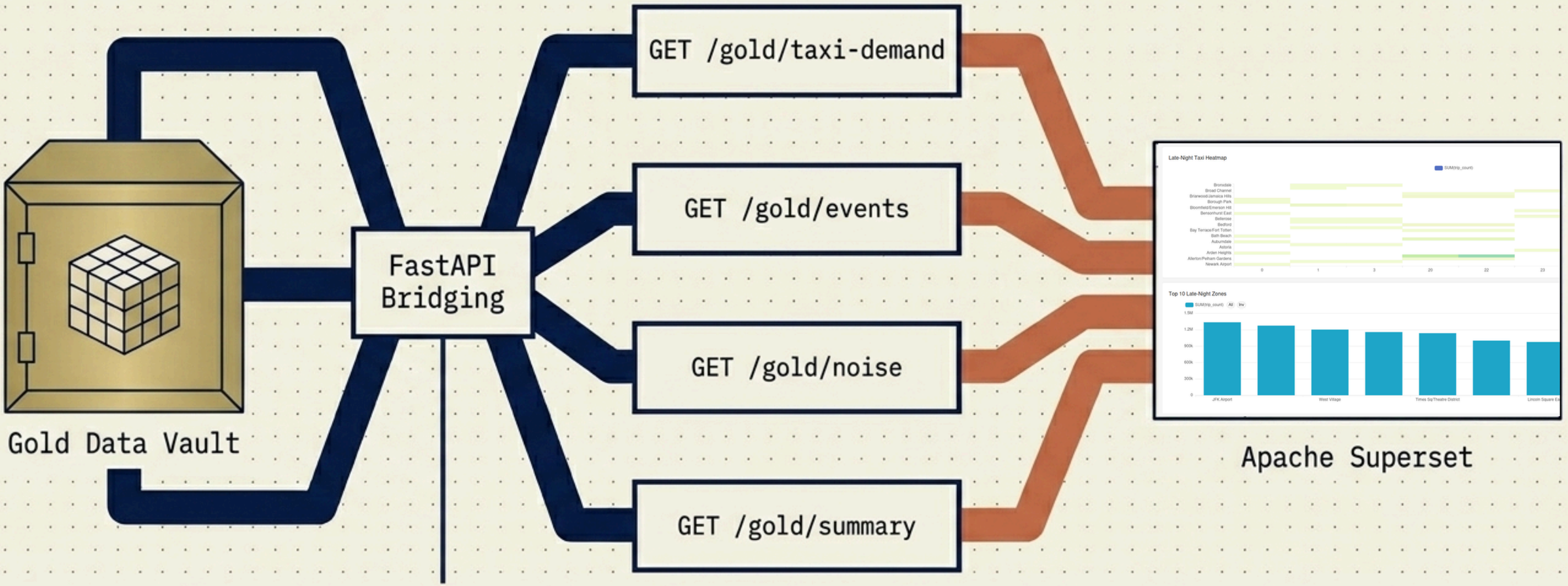


UTC parsing, filtering negative numerics, capping outliers, deriving `trip_duration_minutes` & `is_rush_hour`.

Mandatory column validation, deterministic `event_id` generation via hashing, snapshot deduplication.

Bounding box spatial filtering for NYC, deduplication via `unique_key`, borough normalization.

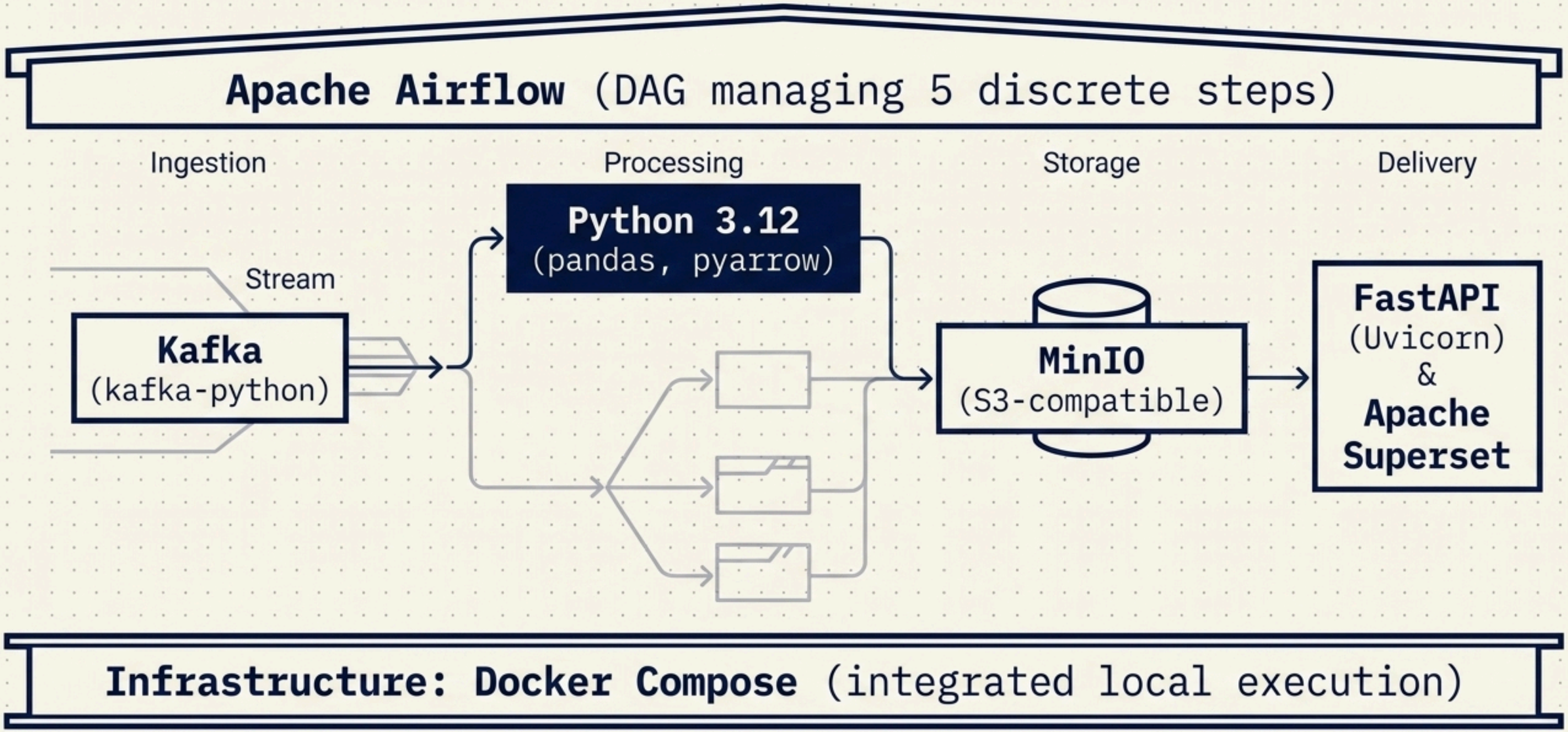
Serving the Pulse: API & Analytics Delivery



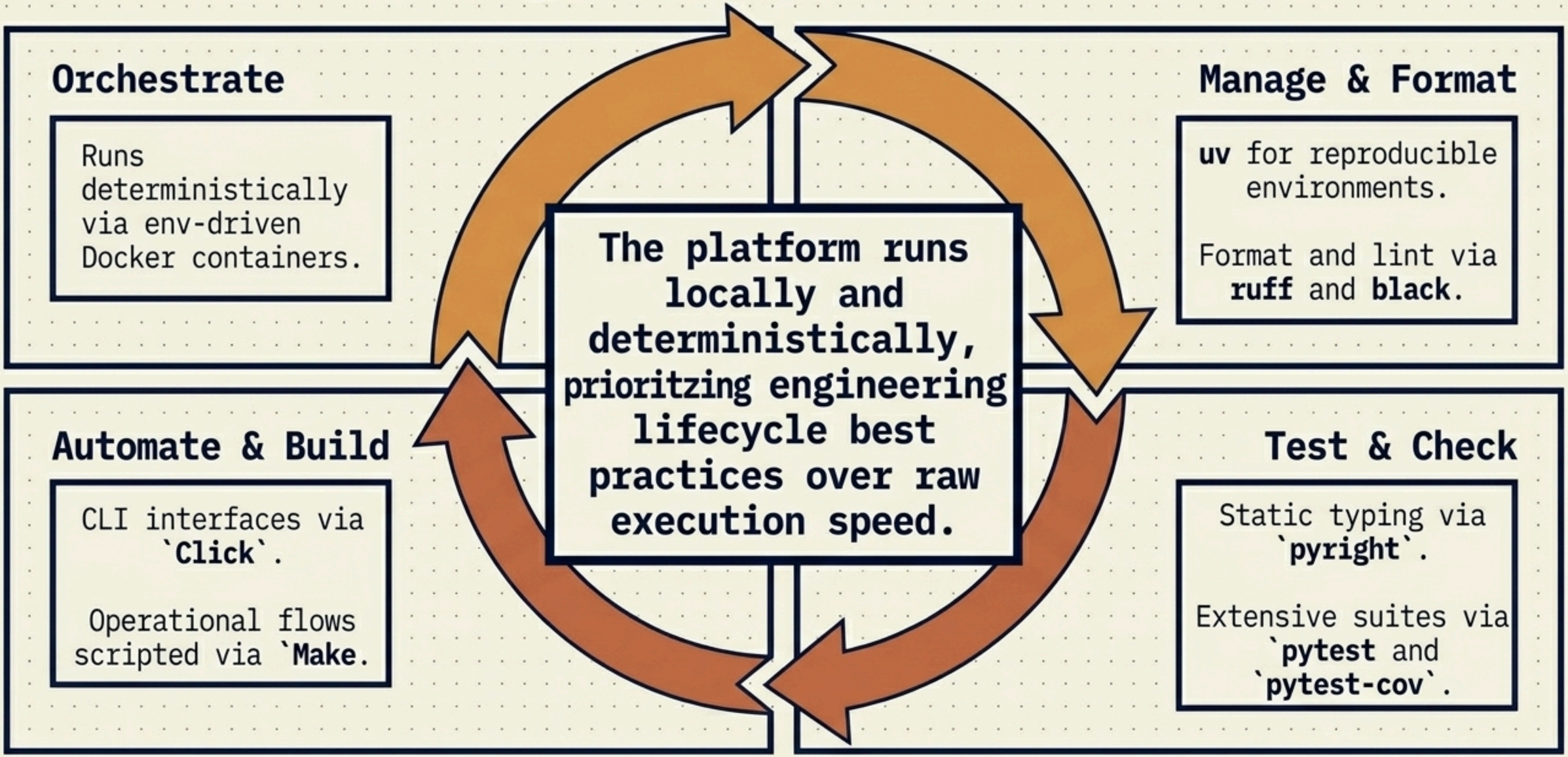
Creates a stable consumption interface, serializing DataFrames safely to JSON.

Consumer Enablement: Built-in query parameters filter by dimension (borough, hour, weekend flags) for visual exploration.

Ecosystem Synthesis: The Technology Stack



Operational Rigor & Reproducibility



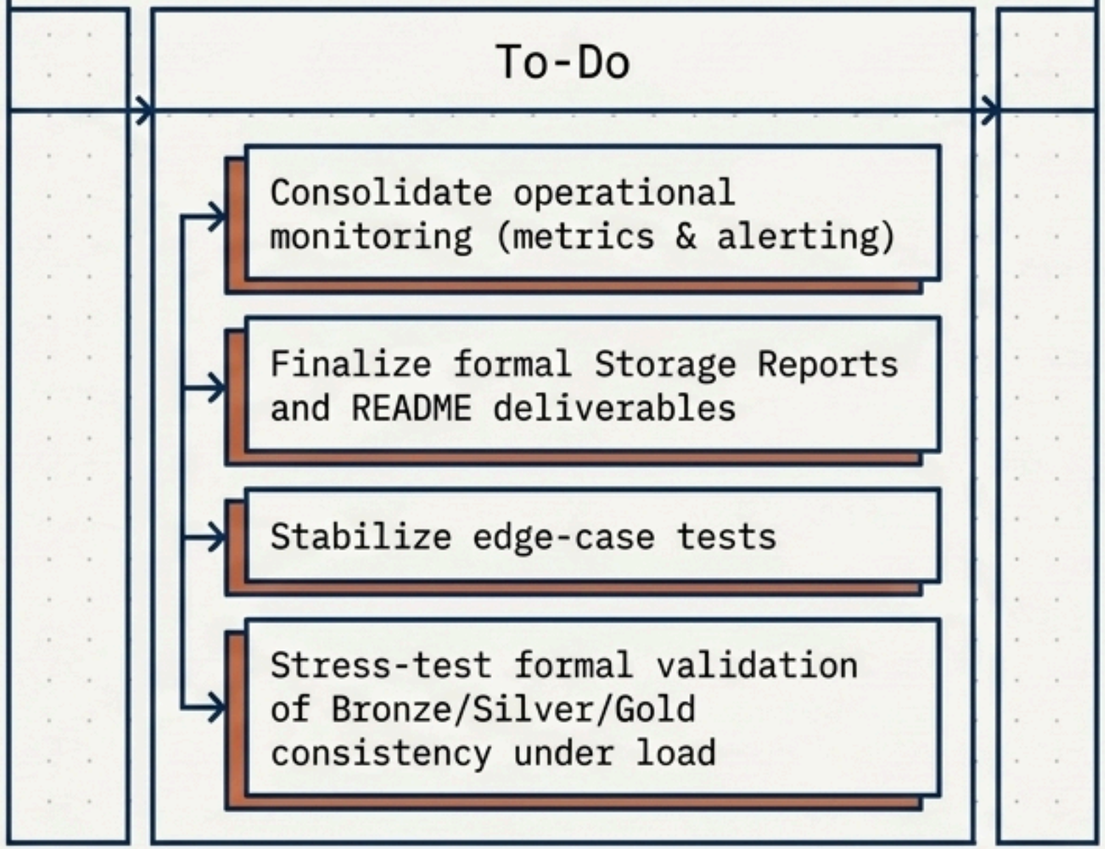
Current State & Forward Roadmap

Takeaway: A mature, transparent engineering lifecycle ensuring the NYC Nightlife Pulse platform is ready for rigorous evaluation.

Current Status (Functional E2E)

- [x] Multi-source ingestion (Batch & Stream)
- [x] Robust stream/batch processing
- [x] Medallion storage persistence
- [x] API serving fully operational

Engineering Backlog



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